Three-dimensional MHD evolution of an ultramassive white dwarf with differential rotation and a mixed field

Results at 192^3 and 256^3 — interim report

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Abstract

We evolve a $2.005\,M_\odot$ equilibrium with strong differential rotation and a toroidal-dominated field in three-dimensional MHD, at two resolutions. The central result is a separation: **the star survives and keeps its mass while its ordered magnetic field is destroyed**. Over 12 s the magnetic energy falls by 95% and the toroidal-to-poloidal ratio drops from 3×10^5 to 10, while the mass varies by 0.42% with no trend. The instability responsible is non-axisymmetric, dominated by $m = 1$, and grows in a few Alfvén times. Its growth rate is *not* converged: it rises by 47% from 192^3 to 256^3, so the measured value is a lower bound. Two runs are still in progress, the 192^3 one already past $t = 22$ s.

1 What is being simulated

1.1 The configuration

The object is a white dwarf of $M = 2.005\,M_\odot$, that is **38% above the Chandrasekhar limit** for $\mu_e = 2$ ($1.456\,M_\odot$). The excess is held up by differential rotation, and that is the physical reason the star is so flattened: $R_{\rm eq} = 3.917 \times 10^8$ cm against $R_{\rm pol} = 1.501 \times 10^8$ cm, or $R_{\rm pol}/R_{\rm eq} = 0.383$.

Mass	$2.0052\,M_\odot$	$1.38\,M_{\rm Ch}$
Central density	3.0×10^9 g cm^{-3}	
Radii	$R_{\rm eq} = 3.917 \times 10^8$ cm	$R_{\rm pol} = 1.501 \times 10^8$ cm
Rotation	$\Omega_c = 8.193$ rad s^{-1}	j-constant law, $A = R_{\rm eq}$
$T/\lvert W\rvert$	0.0993	mass-shedding parameter 0.469
Toroidal field	$\max \lvert B_\varphi\rvert = 3.23 \times 10^{13}$ G	$0.732\,B_c$
Poloidal field	surface dipole 1.0×10^9 G	
Amplitude ratio	$B_t/B_p = 4234$	$E_{\rm tor}/E_{\rm mag} = 0.99999995$
Timescales	$t_{\rm dyn} = 0.475$ s	$t_A = 2.383$ s, $P_{\rm rot} = 0.767$ s

Table 1: Properties of the initial model `rotating_mixed.txt`.

The rotation law is j-constant, $\Omega(\varpi) = \Omega_c A^2/(A^2 + \varpi^2)$ with $A = R_{\rm eq}$. It is the most differential case of that family and leaves the star *marginally* stable by the Rayleigh criterion ($\mathrm{d}j/\mathrm{d}\varpi = 0$).

1.2 The fields

The field is **mixed but toroidal-dominated to one part in 10^7 by energy**. The poloidal component exists as a weak 10^9 G surface dipole; integrated, $E_{\rm pol}/|W| = 5.2 \times 10^{-10}$. This is the collaboration's specification — strong toroidal interior, weak exterior dipole — and it is precisely the combination that makes a nearly purely toroidal field Tayler-unstable.

1.3 Equation of state

`ztwd`: a **zero-temperature, fully degenerate** electron gas with $\mu_e = 2$. It is **barotropic**, $P = P(\rho)$: all temperature derivatives vanish identically and the heat capacities are zero. Two consequences follow, taken up in Sec. 4 — there is no thermal reservoir to receive dissipated energy, and the stratification is neutral, with no buoyancy to limit magnetic instabilities.

1.4 Numerical method

- **Code:** Castro 26.07 on AMReX; ideal MHD with constrained transport, so $\operatorname{div} \mathbf{B}$ is preserved by construction; HLLD Riemann solver; PPM off.
- **Grid:** 3D Cartesian, single level — Castro's constrained transport does not support AMR. Box of 1.8×10^9 cm per side, identical at both resolutions.
- **Resolutions:** 192^3 ($\mathrm{d}x = 9.375 \times 10^6$ cm) and 256^3 (7.031×10^6 cm).
- **Gravity:** `PoissonGrav`, full 3D Poisson solve. Not monopole — with $R_{\mathrm{pol}}/R_{\mathrm{eq}} = 0.38$ the quadrupole is large and a monopole approximation makes the star ring.
- **Initial condition:** the axisymmetric model is built by Grad–Shafranov/SCF on a 385×769 (ϖ, z) grid and interpolated onto the Cartesian mesh from the vector potential. The reconstruction retains 96.0% of the peak $|B_\varphi|$ at 96^3 and 98.9% at 192^3. The field enters in Heaviside–Lorentz units, $\mathbf{B}' = \mathbf{B}/\sqrt{4\pi}$.
- **Robustness:** density floor $10^2\,\mathrm{g\,cm^{-3}}$ against an ambient of 2.0×10^4; sponge between 10^4 and 10^5; velocity damping over the first $2\,t_{\mathrm{dyn}}$, targeted at the *model's* velocity field rather than at zero.
- **CFL:** 0.5 at 192^3; 0.3 at 256^3 from $t \simeq 4.8$ s, needed to get through saturation.

1.5 Diagnostics

Castro's native magnetic diagnostics (`emag_density`, `etor_density`, `Div_B`) proved unusable: all three pack face-centred components of different index types into a single cell-centred FAB and read past the end of the data box, with `Div_B` reaching 3×10^{146} where it should be ~ 0. The components B_x, B_y, B_z are written correctly, and every number here was rebuilt from them with purpose-written tools (`fbtbp`, `fmodes`, `fslice`), using

$$B_\varphi = \frac{-y\,B_x + x\,B_y}{\varpi}, \qquad B_{\mathrm{pol}}^2 = B_x^2 + B_y^2 + B_z^2 - B_\varphi^2.$$

Mass and radii are integrated over cells with $\rho > 10^5\,\mathrm{g\,cm^{-3}}$, so they refer to the *star* and not to the box.

2 Results

2.1 The star survives; the field does not

This is the central result and it holds at both resolutions.

	$t=0$	$t=12\,\mathrm{s}$	ratio
Stellar mass ($M_\odot$)	2.00667	2.00160	0.9975
E_{tor} (erg)	6.035×10^{49}	2.947×10^{48}	**0.049**
E_{mag} (erg)	6.035×10^{49}	3.078×10^{48}	0.051
$E_{\mathrm{tor}}/E_{\mathrm{pol}}$	3.11×10^{5}	22.6	
$\max\lvert B_\varphi\rvert$ (G)	3.196×10^{13}	3.539×10^{13}	1.11

Table 2: Initial and final state of the 192^3 run. The mass is conserved to 0.42% in amplitude over the whole series, with no trend, while the magnetic energy falls by a factor of 20.

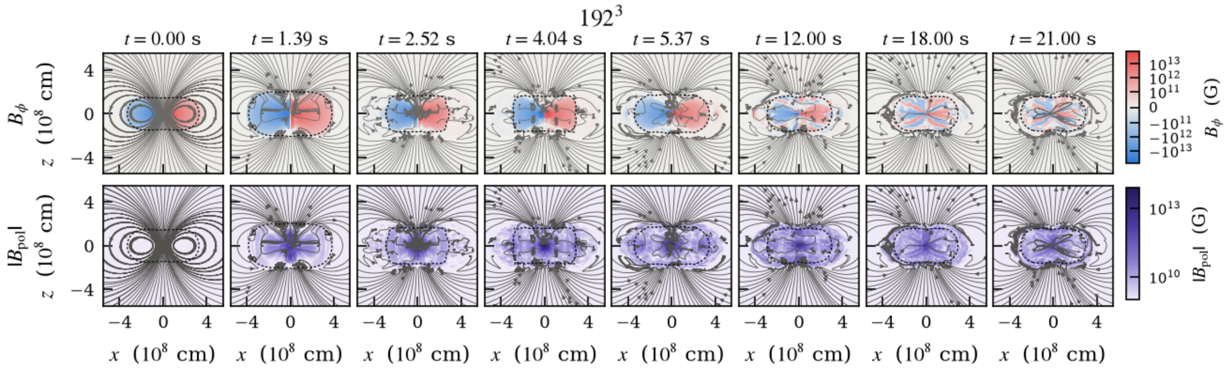

Figure 1: 192^3: toroidal field (top) and poloidal magnitude (bottom) in the meridional plane $y=0$, where B_y *is* B_φ and (B_x,B_z) *is* the poloidal field. Lines: poloidal field, integrated from $\mathbf{B}$. Dashed: $\rho=10^6\,\mathrm{g\,cm^{-3}}$. At $t=0$ the two lobes of opposite sign are the antisymmetry $B_\varphi(x)=-B_\varphi(-x)$. Colour scale fixed across all panels and equal to that of Fig. 6.

Note the last line: **the peak field does not fall — it rises by 11%**. What disappears is the large-scale organisation, not the local strength. Defining an effective volume by $V_{\mathrm{eff}} = 8\pi E_{\mathrm{tor}}/\max|B_\varphi|^2$, the fraction of the star occupied by the toroidal field falls from 1.23% to 0.066%: a factor of 18 in concentration. The field becomes filamentary.

Fig. 1 shows the process. The toroidal field starts as two clean lobes, develops radial striations by $t\simeq 4\,\mathrm{s}$, and ends, between $t=12$ and $21\,\mathrm{s}$, as a fan of thin filaments alternating in sign. The poloidal field does the opposite: it starts as a clean dipole carrying 10^{-7} of the magnetic energy and ends up filling the star, its lines tangled beyond any flux function.

2.2 Density, mass, radius and volume

The mass stays between 1.99971 and 2.00822 $M_\odot$ — 0.42% in amplitude, no trend — and converges towards the model value, 2.00515, as the ambient material captured by the cut disperses. **The star loses no matter**, despite losing 95% of its magnetic energy.

What it does instead is pulsate, hard: the volume swings by a factor of 3.46 over the run, peaking at $2.51\,V_0$ at $t=1.8\,\mathrm{s}$. The mean density runs from 1.31 to $4.53\times10^7\,\mathrm{g\,cm^{-3}}$. By $t=12\,\mathrm{s}$ the star is *smaller* than it began ($0.74\,V_0$) but with half its initial central density: it has become markedly **less centrally concentrated**, with $\rho_c/\langle\rho\rangle$ falling from 91 to 34.

Past the initial transient the pulsation **does not damp**: it locks into a clean periodic mode (Fig. 3a) with

$$P = 1.498\ \mathrm{s} = 3.15\,t_{\mathrm{dyn}} = 1.95\,P_{\mathrm{rot}}, \qquad \text{amplitude } \pm 14.4\% \text{ in } \rho_{\max},$$

its envelope decaying with an e-folding near 140 s. Successive peaks: 1.531, 1.530, 1.540, 1.528, 1.530, 1.532 ($\times10^9$).

2.3 Evolution of the fields

The field goes through three distinct phases (Fig. 4):

Growth, $t = 1.3$–$4.0\,\mathrm{s}$. The poloidal energy grows exponentially, $\sigma = 1.369\,\mathrm{s}^{-1}$ with $r = 0.994$, starting from 1.94×10^{44} erg — which is the interpolation floor, not physics. The e-folding of the *amplitude* is $1.46\,\mathrm{s} = 0.61\,t_A$.

Saturation, $t \simeq 5.3\,\mathrm{s}$. E_{pol} reaches 1.88×10^{48} erg, nearly 10^4 times the floor. The amplitude ratio $\max|B_\varphi|/\max|B_{\mathrm{pol}}|$ falls to 1.19: the poloidal field catches the toroidal one in peak strength. In energy it only reaches 10%, because the toroidal field occupies far more volume — the two ratios saturate in different places and quoting either alone misdescribes the state.

Decay, $t > 5.4\,\mathrm{s}$. E_{tor} falls with an e-folding of $3.44\,\mathrm{s} = 1.44\,t_A$. At $t = 12\,\mathrm{s}$, 4.9% of the initial toroidal energy remains.

The instability is $m = 1$. An azimuthal Fourier decomposition (the `fmodes` tool, a direct projection over cells with no interpolation) gives $|m{=}1|/|m{=}0| = 1.2 \times 10^{-16}$ at $t = 0$ — the initial condition is axisymmetric to machine precision. From $t \simeq 2.5\,\mathrm{s}$ **the poloidal field becomes $m = 1$ dominated**, with $|B_{z,m=1}|/|B_{z,m=0}|$ reaching 21 at $t = 4\,\mathrm{s}$, while the toroidal field stays largely axisymmetric ($m1/m0$ climbing only to 0.14). In the final state the toroidal $m = 1$ saturates at $\simeq 10\%$ of the axisymmetric component.

This rules out the axisymmetric Balbus–Hawley MRI, which in any case could not exist on this grid: built from the *vertical* field ($B_{\mathrm{pol,rms}} = 7.6 \times 10^8\,\mathrm{G}$, $v_{A,\mathrm{pol}} = 3.7 \times 10^4\,\mathrm{cm\,s^{-1}}$), the most unstable wavelength is $\lambda_{\mathrm{MRI}} = 2.9 \times 10^4\,\mathrm{cm}$, or 0.003 of a cell. What remains is the Tayler kink or the non-axisymmetric toroidal-field MRI, both $m \geq 1$, which the azimuthal mode number alone cannot separate.

2.4 Angular momentum

L_z falls at $0.206\%\,\mathrm{s}^{-1}$ while the toroidal field lives and at $0.099\%\,\mathrm{s}^{-1}$ afterwards (Fig. 3b) — **a factor of 2.1 slower once 95% of the magnetic energy is gone**. This is direct evidence that the angular momentum transport was magnetic. At the present rate, another 10% of L_z would take $\sim 100\,\mathrm{s}$.

3 Numerical convergence

Three tests, three different verdicts.

Converged: mass and volume. The mass lies in 1.99971–2.00822 $M_\odot$ at 192^3 and 2.00208–2.00730 at 256^3; the volume-equivalent radius differs by 0.03% at $t = 0$. There is no figure to make.

Not converged: the growth rate. This is the point that has to accompany any version of the result:

	192^3	256^3	ratio
σ from the poloidal energy	$1.369\ \mathrm{s}^{-1}$	$2.007\ \mathrm{s}^{-1}$	1.47
σ from the $m = 1$ mode of B_φ	$1.968\ \mathrm{s}^{-1}$	$2.752\ \mathrm{s}^{-1}$	1.40
$E_{\mathrm{pol}}(t = 0)$, the numerical floor	1.94×10^{44}	1.16×10^{44}	0.60
$\min(\max \lvert B_\varphi \rvert / \max \lvert B_{\mathrm{pol}} \rvert)$	1.19	**0.805**	

The rate *rises* by 47% with resolution, measured by two independent diagnostics that agree on the factor, and the finer grid starts from a *smaller* seed. It also carries the instability further: the poloidal field overtakes the toroidal in peak strength at 256^3 and never does at 192^3. **The measured σ is a lower bound**, and so is the strength of the instability.

Numerical: the early pulsation. The first volume peak falls from $2.51\,V_0$ to 1.79 and the second from an amplitude of 0.49 to 0.11 (Fig. 5b). The breathing that dominates the first few seconds is largely the initial condition ringing on the grid. The persistent 1.50 s mode that appears after $t = 12$ s is a different phenomenon and has not yet been tested against resolution.

A note on cost: settling the rate would take 384^3, about five times the cost of 256^3. The alternative — and probably the right one — is to introduce explicit resistivity, giving the problem a physical dissipation scale instead of letting the mesh set it. CASTRO does not currently implement it.

4 Caveats

1. **The EOS is barotropic.** With `ztwd` there is no thermal reservoir: the 5.7×10^{49} erg that leave the field have no physical destination and are dissipated at the grid scale. We can state that the ordered field is destroyed; we cannot say where the energy went, nor how much heating it would cause in a real star. The stratification is also neutral, with no buoyancy to limit the mode — in a stratified star the rate would be lower, which makes σ a *physical* upper bound at the same time as it is a *numerical* lower bound.
2. **Part of the run lies outside the EOS validity range.** The peak field reaches $1.41\,B_c$ at $t = 4.0$ s and exceeds the Landau critical field in 44% of the sampled instants. `ztwd` assumes an unquantised gas.
3. **Tayler versus toroidal MRI is unresolved.** Both are $m \geq 1$. The non-rotating case from the companion work does not decide it: there the star collapses radially in 2.5 s, through the axial column where B_φ vanishes, and the non-axisymmetric mode is never reached. Set against that, the present result is that **rotation trades one fate for another**: the star stops collapsing and starts losing its field.

5 Status of the simulations

- **Extended** 192^3 (`wdrot192long`): running, already past $t = 22$ s, target 60 s. The time step has quadrupled since $t = 12$ s as the field died, so it should finish well inside the nine windows budgeted.
- 256^3 (`wdrot256`): running, $t \simeq 7.3$ s of 12. With CFL 0.3 it crossed $t = 5.146$ s, where two earlier attempts aborted deterministically.

Everything — inputs, submission scripts, diagnostic tools, data and figure code — is under version control in the `WD_MAG` repository.

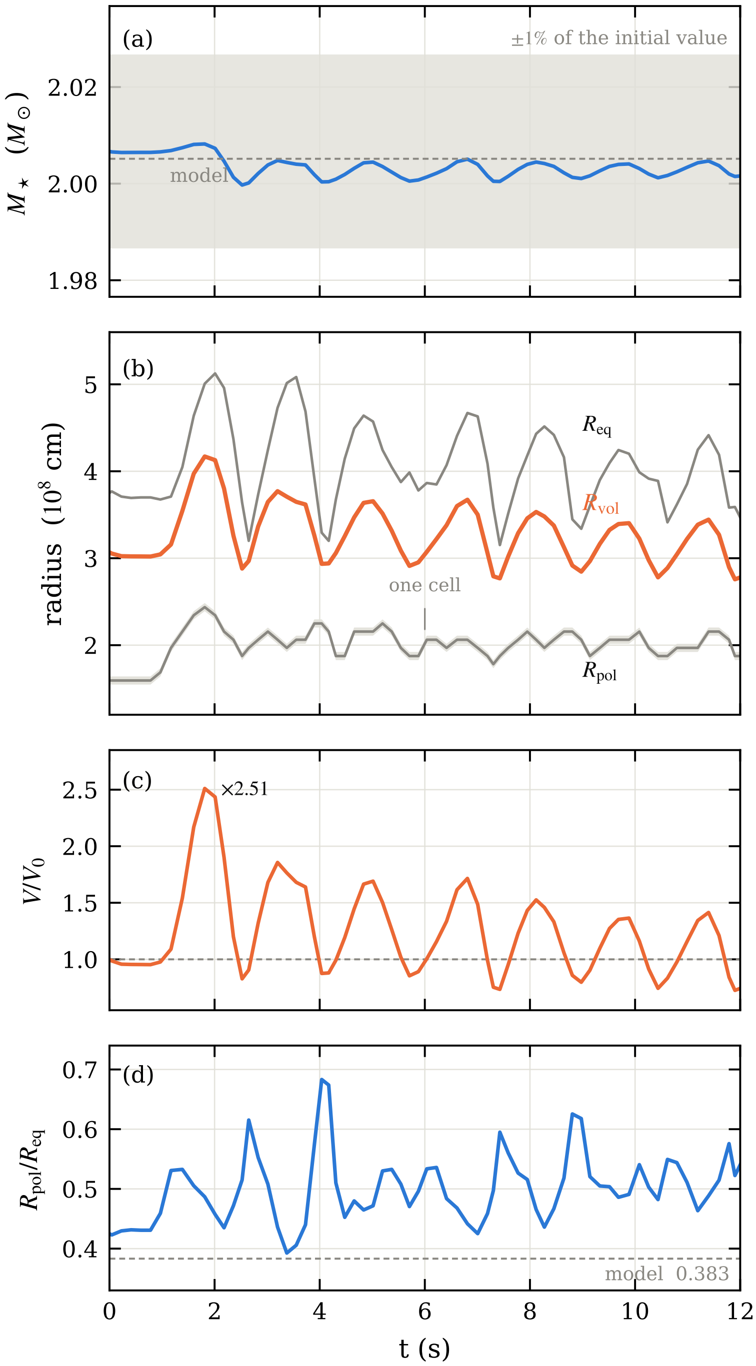

Figure 2: 192^3 to $t = 12\,$s. (a) stellar mass against a ±1% band; (b) the three radii, with one grid cell drawn as a band around R_{pol}; (c) volume; (d) oblateness. The offset in R_{pol}/R_{eq} at $t = 0$ — 0.425 measured against 0.383 in the model — is discretisation, not evolution: R_{pol} quantises onto multiples of dx and R_{eq} is cut short by the density threshold biting into a shallow equatorial profile.

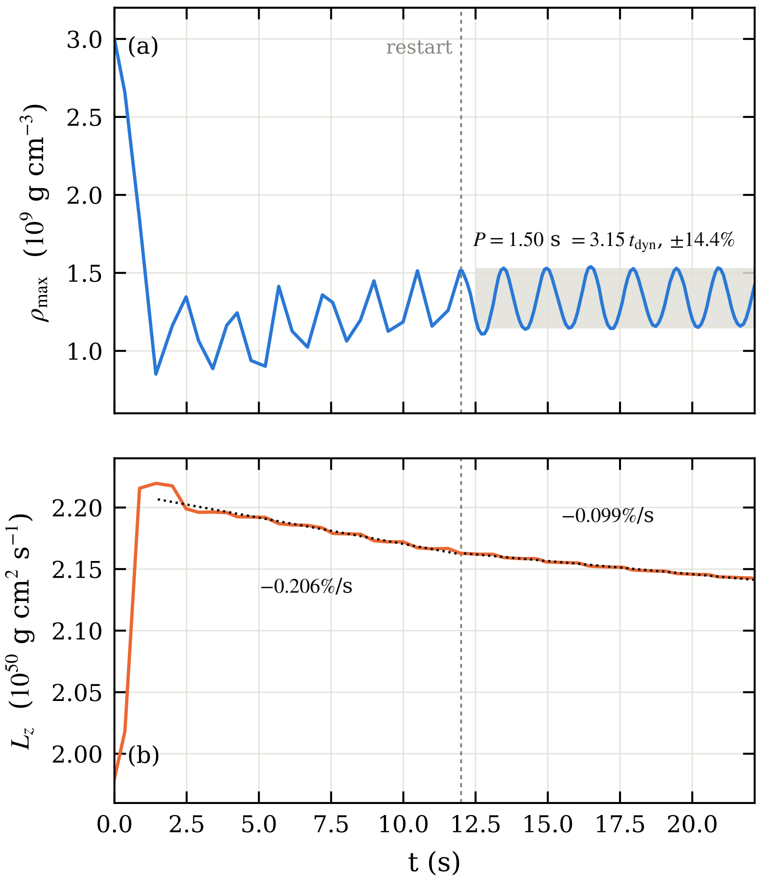

Figure 3: 192^3 extended to $t = 22\,$s (run in progress). (a) central density; (b) axial angular momentum, with the two fitted braking rates. The dashed vertical marks the restart at $t = 12\,$s from `chk03436`; it is a restart, not a discontinuity.

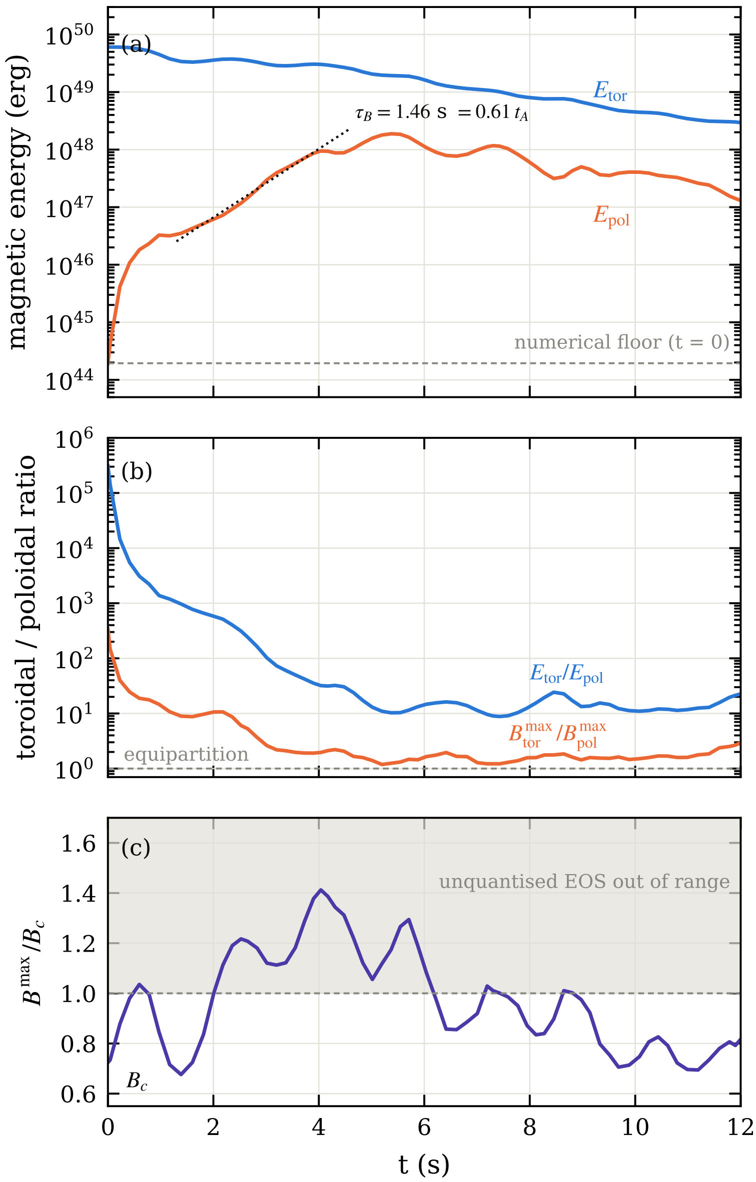

Figure 4: 192^3 to $t = 12\,$s. (a) toroidal and poloidal energies, with the $t = 0$ numerical floor of $E_{\rm pol}$ marked and the exponential fit to the growth phase; (b) the two toroidal-to-poloidal ratios, in energy and in peak amplitude, which saturate in different places; (c) peak field against the Landau critical field.

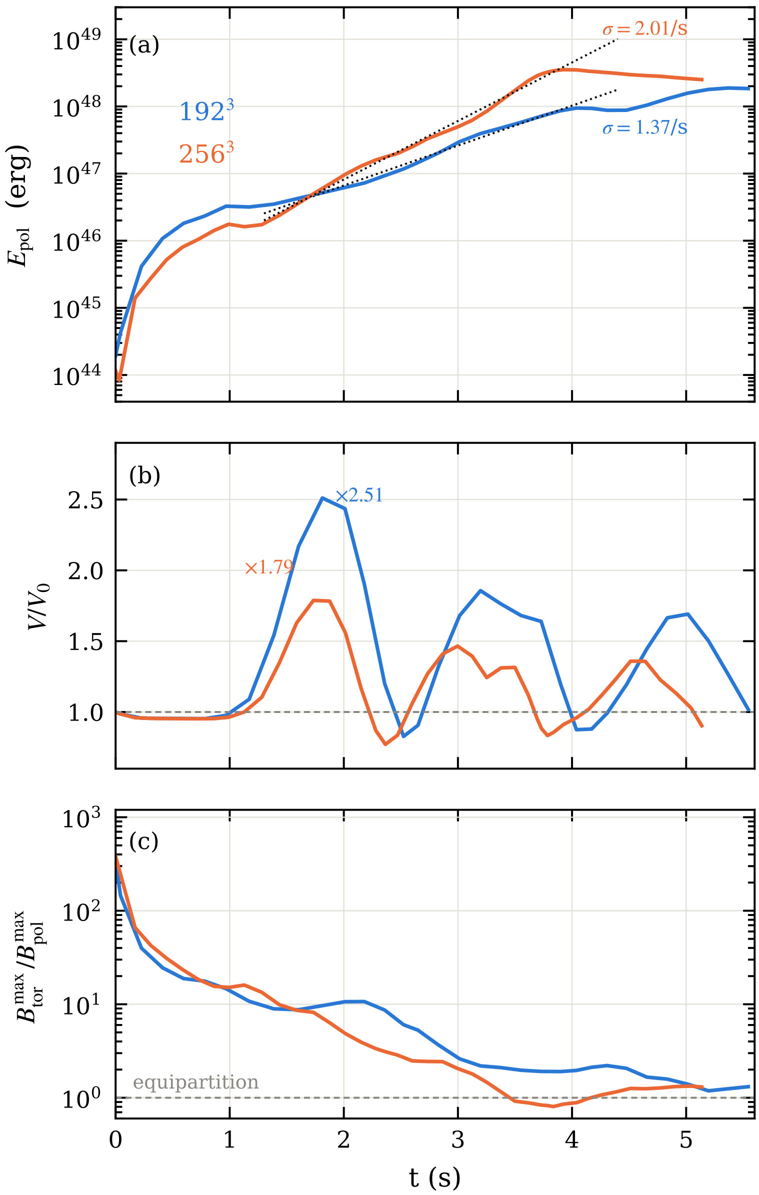

Figure 5: 192^3 against 256^3 over the growth phase. (a) poloidal energy with the fits; (b) volume; (c) amplitude ratio with equipartition marked.

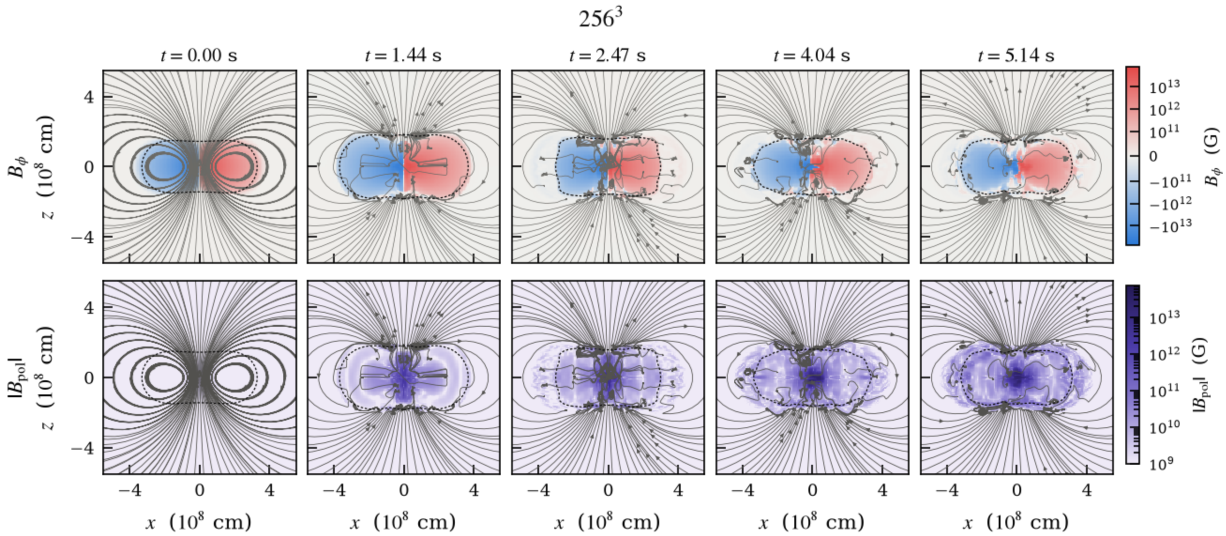

Figure 6: 256^3, same layout and same colour scale as Fig. 1, at the corresponding instants.